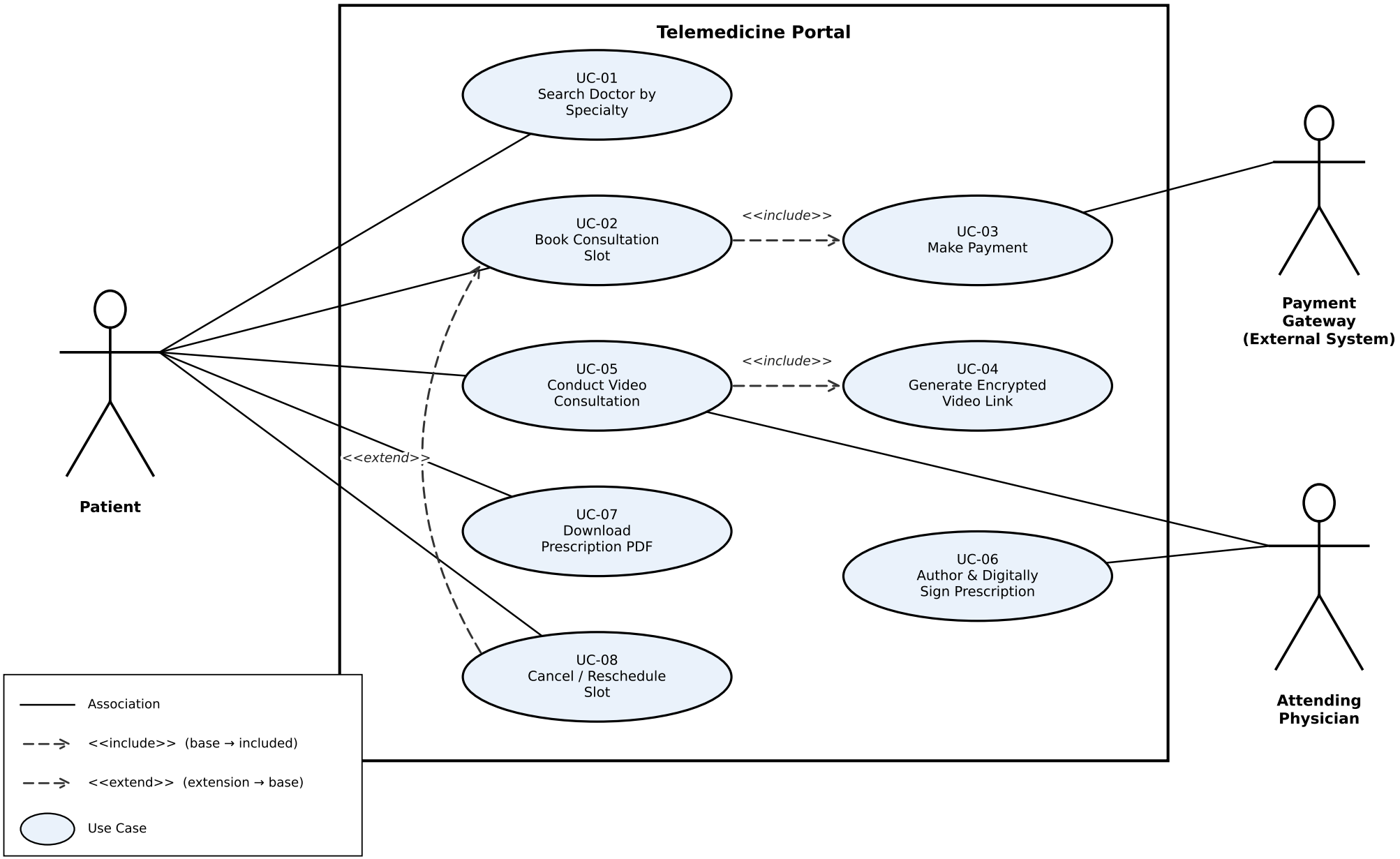
Use-Case Diagram: Telemedicine Slot Booking & Prescription Portal



Complete Use-Case Flow Specification

Telemedicine Slot Booking & Prescription Portal

Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #11

Actors

- Patient — a remote user who searches for physicians, books and pays for consultations, attends the video session, and retrieves prescriptions.
- Attending Physician — a licensed doctor who conducts the consultation and authors and digitally signs prescriptions.
- Payment Gateway — an external system that authorizes consultation payments and processes refunds.

System-Wide Preconditions

- All actors are registered, identity-verified, and authenticated before any use case begins.
- All clinical data is encrypted in transit with TLS 1.3 and at rest with AES-256 (NFR-001).
- The portal is available and durable to the level required by NFR-003.

End-to-End Patient Journey

The eight use cases chain together into one continuous journey. Steps 1–3 form the booking phase, steps 4–5 the consultation phase, and steps 6–7 the prescription phase; step 8 may interrupt the journey at any point before the consultation begins.

1. UC-01 — The patient searches for a physician by specialty and picks one with an open slot.
2. UC-02 — The patient selects a slot; the system holds it and always invokes payment.
3. UC-03 — The Payment Gateway authorizes the fee, converting the hold into a firm reservation.
4. UC-04 — Ten minutes before the appointment, the system issues single-use encrypted room links to both parties.
5. UC-05 — Patient and physician meet in the encrypted video room; clinical notes are captured.
6. UC-06 — If clinically indicated, the physician authors and digitally signs a prescription.
7. UC-07 — The patient downloads the signed prescription PDF for pharmacy fulfilment.
8. UC-08 — At any point before the consultation starts, the patient may cancel or reschedule, triggering slot release and refund.

UC-01 — Search Doctor by Specialty

Actors: Patient (primary)

Preconditions

- Patient is registered and authenticated on the portal.
- The physician directory and slot-availability service are online.

Postconditions

- Success: A filtered list of physicians with genuine open slots is displayed and one is selected.
- Failure: No physician matches the filters; the patient is offered ways to widen the search.

Main Success Scenario

1. Patient opens “Find a Doctor” from the dashboard.
2. System displays the list of medical specialties supported by the portal.
3. Patient selects a specialty and optionally applies filters for date range, consultation language, or fee band.
4. System queries the availability service for physicians matching the filters.
5. System displays each matching physician with qualifications, consultation fee, rating, and next available slot, listing only physicians with at least one open slot in the next seven days.
6. Patient selects a physician to view the full booking calendar.
7. Use case ends successfully and control passes to UC-02.

Alternate Flow — 5a. No Physician Matches the Filters

a. 5a1. System displays a “no availability” message stating which filters were applied.

b. 5a2. System offers to widen the date range, relax the fee band, or show physicians in an adjacent specialty.

c. 5a3. Patient may accept a widened search, or opt in to a notification when a matching slot opens; the use case ends without a selection.

UC-02 — Book Consultation Slot *«include» UC-03: Make Payment*

Actors: Patient (primary)

Preconditions

- Patient has selected a physician via UC-01.
- At least one open consultation slot exists for that physician.
- The patient's account is in good standing with no unpaid dues.

Postconditions

- Success: The slot is reserved exclusively for the patient, payment is captured, and a confirmation with a booking reference is issued.
- Failure: No slot is reserved and no payment is captured; the patient returns to slot selection.

Main Success Scenario

1. Patient selects an available date and time slot from the physician's booking calendar.
2. System displays a booking summary showing physician, specialty, date, time, duration, and consultation fee.
3. Patient confirms the booking request.
4. System places a temporary hold on the slot, valid for 10 minutes, and invokes UC-03 (Make Payment).
5. Payment Gateway authorizes the payment and the system records the transaction reference.
6. System converts the temporary hold into a permanent reservation and removes the slot from all other patients' availability lists.
7. System generates a booking reference and sends confirmation to the patient by email and SMS.
8. System notifies the attending physician and schedules encrypted video-link generation (UC-04).
9. Use case ends successfully.

Alternate Flow — 5a. Payment Declined

a. 5a1. System displays a payment-failed message and warns that the 10-minute hold is still running.

b. 5a2. Patient is prompted to retry with the same or a different payment method.

c. 5a3. If payment fails again after two total attempts, the hold is released, the slot returns to the available pool, and the patient is notified that the booking was cancelled.

Alternate Flow — 4a. Hold Expires Before Payment Completes

a. 4a1. System detects the 10-minute hold has lapsed with no authorization received.
b. 4a2. System releases the slot and voids any pending authorization so the patient is not charged.
c. 4a3. Patient is returned to the calendar and told the slot must be re-selected.

UC-03 — Make Payment *included by UC-02*

Actors: Patient (primary), Payment Gateway (supporting)

Preconditions

- A consultation slot is under temporary hold for this patient.
- The consultation fee has been computed and displayed.

Postconditions

- Success: Payment is captured and a transaction reference is stored against the booking.
- Failure: No funds are captured and the booking is left unpaid.

Main Success Scenario

1. System displays the fee breakdown including consultation fee, platform charges, and applicable taxes.
2. Patient selects a payment method (card, UPI, net banking, or wallet).
3. System securely redirects the patient to the Payment Gateway with the order reference.
4. Patient completes gateway authentication such as OTP or 3-D Secure.
5. Gateway authorizes the payment and returns a signed success response.
6. System verifies the response signature and records the transaction reference against the booking.
7. System displays a payment receipt and emails a copy to the patient.
8. Use case ends successfully.

Alternate Flow — 5a. Gateway Timeout or Ambiguous Response

a. 5a1. System receives no response, or an inconclusive one, within the timeout window.
b. 5a2. System marks the payment “pending” and polls the gateway status endpoint rather than re-charging.
c. 5a3. If the gateway later confirms success, the booking is completed automatically and the patient is notified; if it confirms failure, the hold is released and the patient is invited to retry. Under no circumstances is the patient charged twice.

UC-04 — Generate Encrypted Video Link *included by UC-05*

Actors: System (time-triggered); Patient and Attending Physician as recipients

Preconditions

- A confirmed, paid booking exists for the upcoming slot.
- The scheduled start time is 10 minutes away or less.
- Both parties have a verified contact channel on file.

Postconditions

- Success: A unique, single-use encrypted room link is issued to both parties and the issuance is logged.
- Failure: No usable link is delivered; support is alerted and the session is flagged at risk.

Main Success Scenario

1. Scheduler triggers link generation 10 minutes before the appointment start time.
2. System provisions a video room with a unique room identifier.
3. System issues a separate single-use access token to the patient and to the physician, each bound to that party's identity and to the appointment time window.
4. System encrypts all session metadata in transit using TLS 1.3 and at rest using AES-256.
5. System delivers each link over the recipient's preferred channel (in-app notification, email, and SMS).
6. System writes an issuance record to the audit log capturing time, recipients, and token identifiers.
7. Use case ends successfully.

Alternate Flow — 5a. Link Delivery Fails

a. 5a1. System detects a delivery failure on the primary channel, such as a bounced email.
b. 5a2. System immediately retries on the alternate channel and raises an in-app alert.

c. 5a3. If all channels fail, the session is flagged at risk, support is notified to contact the party by phone, and the booking becomes eligible for reschedule or refund.

UC-05 — Conduct Video Consultation *«include» UC-04: Generate Encrypted Video Link*

Actors: Patient and Attending Physician (both primary)

Preconditions

- A confirmed, paid booking exists for the current time slot.
- Both parties hold a valid single-use encrypted room link (UC-04).
- Both endpoints meet minimum bandwidth and have granted camera and microphone permission.

Postconditions

- Success: The consultation is completed; duration and encrypted clinical notes are stored and the session is marked ready for prescription authoring.
- Failure: The session is marked incomplete with a logged reason and queued for reschedule or refund review.

Main Success Scenario

1. System generates and issues the single-use encrypted room link (UC-04) at least 10 minutes before the scheduled start.
2. Patient and physician each open their link; the system validates the token and the appointment window.
3. System admits the patient to a virtual waiting room and notifies the physician.
4. Physician admits the patient and the encrypted video session begins.
5. System records the session start time and streams audio and video over the secured channel.
6. Physician records clinical notes in the structured consultation form during the session.
7. Either party ends the session; the system records the end time and total duration.
8. System stores the encrypted clinical notes and marks the consultation complete, enabling UC-06.
9. Use case ends successfully.

Alternate Flow — 5a. Connection Drops Mid-Session

a. 5a1. System detects loss of the media stream for one party and pauses the session timer.
b. 5a2. The affected party may rejoin using the same single-use link for up to 10 minutes.
c. 5a3. If neither party rejoins within 10 minutes, the system ends the session, marks it incomplete with the recorded reason, and queues it for reschedule or refund review.

Alternate Flow — 3a. Patient Does Not Join

a. 3a1. System detects the patient has not entered the waiting room by 10 minutes past the start time.
b. 3a2. System sends a reminder to the patient and informs the physician of the no-show timer.
c. 3a3. If the patient has still not joined at 15 minutes, the physician may close the session; it is recorded as a patient no-show and the clinic's no-show policy determines any refund.

UC-06 — Author & Digitally Sign Prescription *«extend» UC-05 (optional outcome of a consultation)*

Actors: Attending Physician (primary)

Preconditions

- A consultation for this patient has been completed and marked complete by the system.
- The physician is authenticated and holds a valid, unexpired medical licence on file.
- The physician's digital signing certificate is active and not revoked.

Postconditions

- Success: A cryptographically signed prescription PDF is stored against the patient record and released for download.
- Failure: No prescription is issued; the draft is retained and the reason is written to the audit log.

Main Success Scenario

1. Physician opens the completed consultation record and selects “Create Prescription”.
2. System displays a structured prescription form pre-filled with patient and consultation details.
3. Physician enters each drug with dosage, frequency, duration, and any special instructions.
4. System validates the entries against the drug formulary and flags dosage or interaction warnings.
5. Physician reviews the warnings, adjusts entries if required, and submits the prescription.

6. System re-validates the physician's licence status and signing certificate.
7. System applies the digital signature and compiles the prescription into a PDF.
8. System stores the signed PDF against the patient record, writes an audit entry, and notifies the patient that it is available.
9. Use case ends successfully.

Alternate Flow — 6a. Licence or Signing Certificate Invalid

a. 6a1. System finds the physician's licence expired or the signing certificate revoked.
b. 6a2. System blocks signing, saves the prescription as an unsigned draft, and displays the specific reason.
c. 6a3. System logs the blocked attempt and alerts the clinic administrator; no PDF is generated and the patient is not notified.

UC-07 — Download Prescription PDF

Actors: Patient (primary)

Preconditions

- A digitally signed prescription exists for the patient's completed consultation.
- Patient is authenticated on the portal.

Postconditions

- Success: The signed PDF is delivered to the patient and the access is recorded in the audit log.
- Failure: The document is withheld and the patient is told why.

Main Success Scenario

1. Patient opens the relevant consultation from “My Consultations”.
2. Patient selects “Download Prescription”.
3. System verifies the patient's identity and confirms they are the subject of that prescription.
4. System verifies that the attached cryptographic signature is intact and the signing certificate was valid at the time of signing.
5. System serves the PDF over a TLS 1.3 connection.
6. System writes an access entry to the audit log capturing patient, document, timestamp, and source address.
7. Use case ends successfully.

Alternate Flow — 4a. Signature Verification Fails

a. 4a1. System finds the signature invalid or the document altered since signing.
b. 4a2. System blocks the download and displays a message that the prescription is undergoing verification.
c. 4a3. System raises a high-priority alert to the clinic administrator and the issuing physician for re-issue; the patient is notified once a valid replacement is available.

UC-08 — Cancel / Reschedule Slot *«extend» UC-02: Book Consultation Slot*

Actors: Patient (primary), Payment Gateway (supporting)

Preconditions

- Patient is authenticated and has at least one confirmed, upcoming booking.
- The consultation has not yet started.
- The cancellation or reschedule window defined by clinic policy is still open.

Postconditions

- Success (cancel): The slot is released, the physician is notified, and any refund due is initiated.
- Success (reschedule): The original slot is released, a new slot is reserved against the same payment, and both parties are notified.
- Failure: The original booking is unchanged and the patient is told why the request was rejected.

Main Success Scenario

1. Patient opens “My Appointments” and selects a confirmed upcoming booking.
2. System displays the booking details together with the applicable cancellation and refund policy.
3. Patient chooses either “Cancel” or “Reschedule”.
4. System validates the request against the clinic's cut-off window.
5. If rescheduling, the patient selects a new available slot for the same physician and the system reserves it against the existing payment.

6. If cancelling, the system releases the slot and initiates a refund through the Payment Gateway at the applicable policy tier.
7. System notifies the attending physician of the change.
8. System sends the patient a confirmation carrying the updated booking reference or the refund reference.
9. Use case ends successfully.

Alternate Flow — 4a. Request Outside the Cut-Off Window

a. 4a1. System detects the request falls inside the no-change window, for example under two hours before the appointment.
b. 4a2. System rejects the change and displays the policy reason together with the non-refundable amount.
c. 4a3. Patient may keep the original booking or cancel without refund; the booking record is updated accordingly and the use case ends.